Pharmaceutical Society of Australia

Hydroxocobalamin (vitamin B12) injection quick guide

Indication

Intramuscular injection of hydroxocobalamin (vitamin B12) is used to treat[1-4]:

- pernicious anaemia and other macrocytic anaemias associated with vitamin B12 deficiency
- vitamin B12 dietary deficiency (used in the initial treatment if the intestinal mucosa has been compromised by a long period of vitamin B12 deficiency causing reversible malabsorption)
- optic neuropathies such as tobacco amblyopia and Leber's optic atrophy
- cyanide poisoning (high doses - hydroxocobalamin 2.5 gram injection).

This quick guide will focus on the use of hydroxocobalamin associated with vitamin B12 deficiency.[1]

Vitamin B12 is essential for nerve development, nucleic acid synthesis and normal erythropoiesis.[1] It is commonly tested for when investigating anaemia and dementia.[2] Early treatment may be essential to prevent neurological damage becoming irreversible. Symptoms present for over a year tend to persist.[3]

Reasons for vitamin B12 deficiency include[1,3,4]:

- impaired absorption (e.g. autoimmune gastritis, bariatric surgery, gastrectomy, gastric bypass, exocrine pancreatic failure, inflammatory bowel disease, loss or disease of the terminal ileum, and intestinal bacterial overgrowth)
- inadequate dietary intake - people who follow a strict vegan diet for a number of years including breastfed infants of mothers who are vegan.

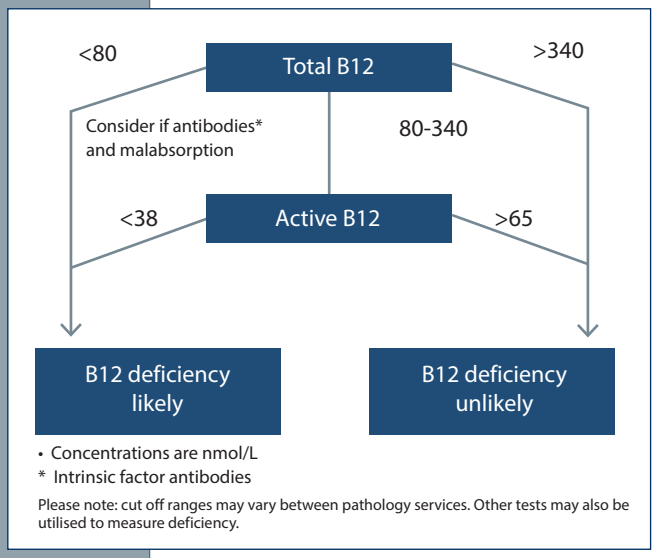
Symptoms of vitamin B12 deficiency

Symptoms of vitamin B12 deficiency include fatigue, lethargy, breathlessness, feeling faint, headaches, pale skin, palpitations, tinnitus, loss of appetite and weight loss, sore and red tongue (glossitis), mouth ulcers, pins and needles (paraesthesia), disturbed vision, irritability, depression, memory loss, tingling in the hands or feet, changes in feeling the hands or feet, and visual disturbances.[5]

Screening and monitoring

The following are some of the more notable precautions and adverse effects. Refer to the product information for the full list of precautions and adverse effects. If you have any concerns, please contact the patient's prescriber before administering the product.

Pathology	Vitamin B12 is present in the circulation bound to two proteins, transcobalamin (known as active B12) and haptocorrin (inactive). If total vitamin B12 is >340 nmol/L, vitamin B12 deficiency unlikely. Up to 30% of patients with a normal total B12, have low levels of active B12, indicating deficiency.[3]
Anaemia	What is the cause of the patient's anaemia? Confirm the cause of anaemia before starting vitamin B12 treatment.
Pregnancy	Is the patient pregnant or breastfeeding? • Vitamin B12 crosses the placental barrier. It is not recommended in pregnancy unless the expected benefits outweigh any potential risk to the infant.[3] • Hydroxocobalamin is distributed into breast milk. It is not recommended for breastfeeding mothers unless the expected benefits to the mother outweigh any potential risk to the infant.[3] Megaloblastic anaemia of pregnancy as usually due to folic acid deficiency rather than vitamin B12 deficiency.
Known allergies	Is the patient allergic to[7-10]: • hydroxocobalamin • any ingredients in the injection • cobalt (e.g. symptoms of an allergic reaction may include shortness of breath, wheezing or difficulty breathing, swelling of the face, lips, tongue or other parts of the body, skin rash, itching or hives). If so, do not inject. Refer the patient to their doctor.



• Concentrations are nmol/L
* Intrinsic factor antibodies

Please note: cut off ranges may vary between pathology services. Other tests may also be utilised to measure deficiency.

Onset of effect	Inform the patient that the onset of effect should be immediate but may take a few days or weeks before symptoms start to improve. Patients with anaemia, malaise will improve within 2 days of first injection.[4] Neuropathy will improve slowly although patients with long history of neurological symptoms may have residual effects.[4]
Initial phase of treatment for pernicious anaemia	Remind the patient that their doctor will monitor their potassium levels during the initial stages of vitamin B12 treatment, and will correct hypokalaemia as needed[1,4]
Monitoring	Remind the patient that their doctor will monitor vitamin B12 levels and they will need to have a full blood count every year[1]
Side effects	Inform the patient of the typical side effects of vitamin B12 injections including: transitory hypertension, diarrhoea, dizziness, headache, nausea, pain/irritation at injection site, vomiting, red-coloured faeces or pink-tinged urine[7-10]
Unable to tolerate IM injections	Is the patient able to tolerate intramuscular vitamin B12? If not, high-dose oral cyanocobalamin (e.g. 1 mg daily) may be tried.[1] Patients will require regular monitoring of their vitamin B12 concentration.

Dose, frequency and expected duration

Condition	Initial dose#	Maintenance dose
• Vitamin B12 malabsorption* • Dietary vitamin B12 deficiency if the intestinal mucosa has been compromised by a long period of B12 deficiency causing reversible malabsorption	Hydroxocobalamin IM 1 mg on alternate days for 1–2 weeks or until improvement occurs#	Hydroxocobalamin IM 1 mg every 2–3 months

References: AMH[1]; MelPath[3]; eTG[4]

Note:

* Pernicious anaemia or bowel resection requires lifelong treatment with vitamin B12 injection.

\# Same dose is used for adults and children.

Administration

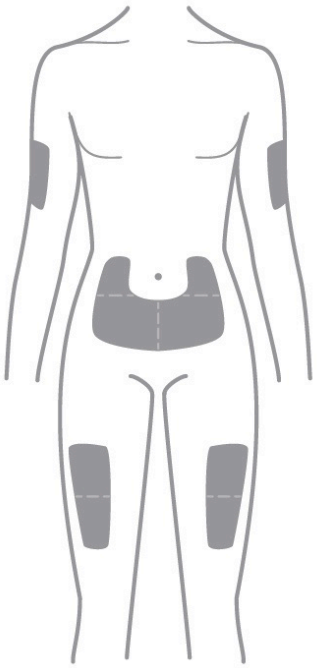
Hydroxocobalamin injection solution 1 mg/mL is either:

- clear, pale red liquid (*Vita-B12, Cobal-B12*) **OR**
- clear, dark red liquid (*Hydroxo-B12, Neo-B12*).

The solution is in a glass ampoule that contains no antimicrobial agent. It is for single use only. Store at room temperature.

Where can hydroxocobalamin be injected?

Hydroxocobalamin is administered **intramuscularly** into either the abdomen, thigh or upper arm.



The choice of injection site may be influenced by patient preference and ease of administration.

How to inject hydroxocobalamin

1. Assemble all necessary equipment to inject the medicine: hydroxocobalamin 1 mg/mL ampoule, 38 mm long 18 gauge drawing up needle and a 25 mm long 22-25 gauge giving needle*, sharps contaminated-waste container, syringe and sterile alcohol swab (*Needle requirements may vary depending on the age and size of the patient. No more than 2 mL of medication should be injected via the intramuscular route).
2. Thoroughly wash and dry your hands.
3. Inspect the ampoule for any colour changes or particulate matter. The solution should be either a clear pale red (e.g. *Vita-B12, Cobal-B12*) or dark red colour (e.g. *Hydroxo-B12, Neo-B12*).[7-10]
4. Tap the ampoule to ensure all the solution is at the bottom of the ampoule.
5. Wipe the top of the glass ampoule with an alcohol swab. Let dry for 30 seconds.
6. With the swab around the neck of the ampoule, snap the top off in a direction that's away from your body.
7. Attach the drawing up needle to the syringe.
8. Remove the needle cap, and place the needle into the bottom of the ampoule.
9. Pull the syringe plunger upwards until all the solution is removed from the ampoule.
10. Remove the drawing up needle from the syringe and place it in the sharps container.
11. Push the plunger slowly down until there is 1 mL of the solution in the syringe ensuring that the solution is at the tip of the syringe.
12. Attach the giving needle to the syringe.
13. Place the needle at a 90 degree angle to the skin.[11,12]
14. Push the syringe plunger down to inject all the solution into the muscle, wait 10 seconds and then carefully remove the needle from the injection site.
15. Do not rub the injection site once the needle has been removed.
16. Place all used injection equipment into the sharps container.

Product

1 mL injection (1 mg/mL). Discard any residue. Store below 25º C (unscheduled).

Products available:

NEO-B12 - Hydroxocobalamin chloride (Pfizer Australia Pty Ltd)

VITA-B12 – Hydroxocobalamin acetate (Amdipharm Mercury (Australia) Pty Ltd)

COBAL-B12 - Hydroxocobalamin acetate (Amdipharm Mercury (Australia) Pty Ltd)

Hydroxo-B12 - Hydroxocobalamin chloride (Aspen Pharmacare Australia Pty Ltd)

References

1. Blood and electrolytes. In: Rossi S, ed. Australian Medicines Handbook; 2025. Online. Adelaide. At: https://amhonline.amh.net.au/chapters/blood-electrolytes/drugs-anaemias/other-drugs-anaemias/vitamin-b12
2. Green R. Vitamin B12 deficiency from the perspective of a practicing hematologist. Blood 2017; 29(19):2603-11.
3. Vitamin B12. Melbourne Pathology. Insight 2022. At: https://www.mps.com.au/media/jx-5b2omr/17353_mp_insight_vitamin-b12_2021-v2.pdf
4. Expert Group for Gastrointestinal. Gastrointestinal, version 6. In eTG complete. Melbourne: Therapeutic Guidelines; 2021.
5. HealthDirect. B12 deficiencies. 2024. At: www.healthdirect.gov.au/b12-deficiencies
6. Lab Tests Online AU. What You Should Know About Your B12 And Folate Tests.2010. https://www.pathologytestsexplained.org.au/content/pdf/1437%20PTEx%20A4%20B12_Folate%20fin.pdf
7. Neo-B12 Injection (Hydroxocobalamin) approved product information. 2022. www.ebs.tga.gov.au/ebs/picmi/picmirepository.nsf/pdf?Opengent&id=CP-2018-PI-01910-1
8. Vita-B12 Injection (Hydroxocobalamin) approved product information. 2022. At: www.ebs.tga.gov.au/ebs/picmi/picmirepository.nsf/pdf?Opengent&id=CP-2012-PI-01726-3
9. Cobal-B12 Injection (Hydroxocobalamin) approved product information. 2022. At: www.ebs.tga.gov.au/ebs/picmi/picmirepository.nsf/pdf?OpenAgent&id=CP-2017-PI-01629-1&d=20220315172310101
10. Hydroxo-B12 Injection (Hydroxocobalamin) approved product information. 2022. At: www.ebs.tga.gov.au/ebs/picmi/picmirepository.nsf/pdf?Opengent&id=CP-2010-PI-07721-3
11. Australian Technical Advisory Group on Immunisation (ATAGI). Australian Immunisation Handbook. Administration of vaccines. 2018. At: https://immunisationhandbook.health.gov.au/vaccination-procedures/administration-of-vaccines
12. NHS Trust. University Hospitals Plymouth. How to Self-Administer a B12 Injection.2020. At: www.livewellsouthwest.co.uk/wp-content/uploads/2020/04/Vitamin-B12-self-admimistration-patient-guide.pdf

PSA5795